

Project Design Phase

Proposed Solution Template

Date	12 September 2026
Team ID	
Project Name	Employee Laptop Request Management System
Maximum Marks	2 Marks

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Employees need a consistent and controlled way to request laptops. Manual email or verbal requests can cause incomplete information, duplicate requests, unclear approval status and limited traceability.
2	Idea / Solution description	Implement a structured Employee Laptop Request Management System in ServiceNow. The solution uses a Record Producer for submission, a custom Laptop Request Form table, Business Rule validation, Flow Designer approval, email notifications, roles and ACLs.
3	Novelty / Uniqueness	The solution connects request submission, annual duplicate prevention, manager approval, status updates, notifications and role-based access into one traceable ServiceNow lifecycle instead of handling each activity separately.
4	Social Impact / Customer Satisfaction	A standardized request process reduces manual follow-up, gives employees clear status information and helps managers make accountable approval decisions.
5	Business Model (Revenue Model)	This is an internal enterprise IT service-management solution. It is not designed as a standalone revenue-generating product. Business value is realized through reduced manual handling, better control and improved operational efficiency.
6	Scalability of the Solution	The scoped ServiceNow application can support additional employees, requests, managers and IT groups. The same request and approval design can be extended to other employee hardware or service requests.

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Problem – Solution Fit Template

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The Problem–Solution Fit below connects employee needs, triggers, behaviour, pain points, expected gains and the proposed ServiceNow solution.

1. CUSTOMER SEGMENTS	2. JOBS-TO-BE-DONE / PROBLEMS	3. TRIGGERS
Employees requesting laptops; IT Managers; IT Employees; Project Team.	Submit a complete laptop request; select Basic/Advance; provide reason; receive approval; track status; prevent duplicate annual requests.	Employee needs a laptop; request must be submitted; duplicate check is required; manager approval is needed; outcome must be communicated.
4. EMOTIONS: BEFORE / AFTER	5. AVAILABLE SOLUTIONS	6. CUSTOMER CONSTRAINTS
Before: uncertainty about the manual process, approval and status. After: greater clarity, confidence and traceability.	Email requests; verbal requests; spreadsheets or informal tracking; manual manager follow-up.	Must use the existing ServiceNow environment; users have different roles; requests and approvals must remain traceable.
7. PAINS / ROOT CAUSES	8. BEHAVIOUR	9. GAINS / EXPECTED BEHAVIOUR
Incomplete request information; duplicate requests; unclear approval state; manual follow-up; unrestricted access can create control issues.	Employee enters request details; system validates the annual policy; manager reviews; status and email outcome are updated.	Standardized request entry; duplicate prevention; controlled approval; clear status; role-based access; better accountability.
10. OUR SOLUTION	11. VALUE / FIT	12. CHANNELS / TOUCHPOINTS
A ServiceNow Employee Laptop Request workflow using a Record Producer, custom table, Business Rule, Flow Designer, approval records, notifications, roles and ACLs.	The solution addresses the main operational pain points at the same stages where they occur while keeping the request and approval lifecycle connected.	ServiceNow Service Portal / Record Producer, Laptop Request Form, Flow Designer, approval record, email notifications and role-based views.

Fit conclusion: The proposed ServiceNow workflow is a strong fit because it replaces informal laptop requests with a controlled submission, validation, approval and notification lifecycle.

Project Design Phase

Solution Architecture

Date	12 September 2026
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Maximum Marks	4 Marks

Solution Architecture:

The architecture connects the employee-facing Record Producer with the custom Laptop Request Form table, backend duplicate validation, Flow Designer approval, manager decision, status updates, email notifications and role-based security. The update set provides controlled deployment of the configured application artifacts.

Solution Architecture - Employee Laptop Request Management System

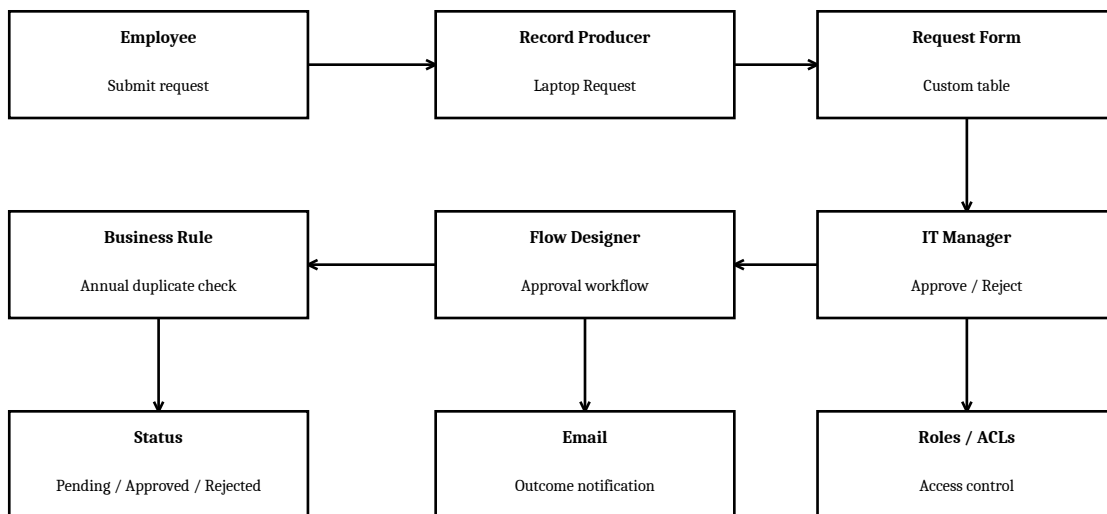


Figure 1: Solution architecture and data flow of the Employee Laptop Request Management System.

Key architectural flow:

- Employee – ServiceNow Record Producer – Laptop Request Form.
- Business Rule checks the requester for an existing request in the current calendar year.
- Flow Designer requests manager approval through the sysapproval_approver relationship.
- Approval outcome updates the request status and triggers an email notification.
- Roles and ACLs control employee and manager access; the update set captures deployment artifacts.

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Solution Architecture

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Architecture Components and Responsibilities

Component	Responsibility
Service Portal / Record Producer	Provides the employee-facing Laptop Request submission interface.
Laptop Request Form Table	Stores requester, email, request date, description, manager, type, status and reason.
Business Rule	Prevents more than one laptop request by the same requester within the current calendar year.
Flow Designer	Sets Pending Approval, requests manager approval, handles Approved/Rejected outcomes and supports notifications.
sysapproval_approver	Stores the manager approval relationship and approval decision.
Roles and ACLs	IT employees can create/read/write requests; IT managers have controlled read access according to the configured ACL design.
Email Notifications	Communicate manager approval requests and final approval/rejection outcomes.
Update Set	Captures the scoped application configuration for preview, commit and deployment.